Grade 11 Chemistry: Chemical Bonding

Chemical bonding is the process by which atoms combine to form compounds. Understanding bonding helps explain the structure, properties, and behavior of matter.

1. Types of Chemical Bonds

There are three main types of chemical bonds:

1. Ionic Bonding – transfer of electrons from one atom to another.
2. Covalent Bonding – sharing of electrons between atoms.
3. Metallic Bonding – electrons are shared in a 'sea of electrons' among metal atoms.

2. Ionic Bonding

Ionic bonds form when electrons are transferred from a metal to a nonmetal, creating ions.

1. Metal loses electrons → forms a positive ion (cation).
2. Nonmetal gains electrons → forms a negative ion (anion).
3. Opposite charges attract to form the bond.
4. Example: Sodium Chloride (NaCl).

3. Covalent Bonding

Covalent bonds occur when atoms share electron pairs.

1. Usually occurs between nonmetals.
2. Can form single, double, or triple bonds.
3. Can be polar or nonpolar depending on electronegativity difference.
4. Example: Water (H■O), Oxygen (O■).

4. Metallic Bonding

Metallic bonding involves a lattice of metal ions surrounded by a sea of delocalized electrons.

1. Electrons move freely.
2. Explains conductivity of metals.
3. Gives metals malleability and ductility.

5. Lewis Structures

Lewis structures show valence electrons and how atoms bond.

1. Dots represent valence electrons.
2. Helps visualize bonding and lone pairs.
3. Follows the octet rule.

6. Octet Rule

Atoms tend to gain, lose, or share electrons to achieve 8 valence electrons.

1. Applies mainly to main-group elements.
2. Exceptions: Hydrogen (2 electrons), expanded octets (e.g., phosphorus).

7. Intermolecular Forces

Intermolecular forces are attractions between molecules.

1. London Dispersion Forces – weakest, present in all molecules.
2. Dipole-Dipole Forces – between polar molecules.
3. Hydrogen Bonding – strong interaction involving H, N, O, or F.

Conclusion

Understanding chemical bonding is essential for explaining how substances form and behave. These concepts are foundational for further study in chemistry.